

Active Disturbance Rejection Consensus Control of Multi-Agent Systems Based on a Novel NESO

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Abstract—Distributed consensus control problems of multiagent systems (MASs) subject to total disturbances are studied in this article, where all agents communicate over a fixed topology with undirected or directed connections. First, a novel nonlinear fast convergence function and a novel nonlinear extended state observer (NNESO) are proposed for MASs to realize real-time estimation of total disturbances. Then, an active disturbance rejection control (ADRC) method based a consensus control protocol with disturbance compensation is presented for MASs to reduce the impact of total disturbances on consensus control of MASs. Furthermore, the stability and convergence of MASs are strictly proved by the Lyapunov stability method, which indicates that MASs under constant disturbances can realize consensus without state errors, and MASs under time-varying disturbances can realize consensus with state errors, which can be converged to an arbitrary and assigned neighborhood. Finally, the advantages of the NNEO are demonstrated by comparing the convergence speed of the nonlinear convergence function of the NNEO with that of the conventional ESOs, and the effectiveness of the consensus control method for MASs on the basis of ADRC is verified by numeric simulations and actual experiments.

Index Terms—Active disturbance rejection control (ADRC), consensus, disturbances, multiagent systems (MASs), nonlinear extended state observer (NESO).

I. INTRODUCTION

DURING recent years, there has been extensive research on the coordinated control of multiagent systems (MASs) due to their capacity for solving these problems that are challenging for single-agent systems, such as the collaboration of multiple

robots [1] and the utilization of distributed resources [2]. One of the most fundamental problems in the coordinated control of MASs is the consensus problem, which means that all the agents must attain identical control objectives, such as the same state values and outputs of MASs [3], [4]. Nowadays, consensus control of MASs is widely applied in various fields, such as unmanned aerial vehicles (UAVs) [5], [6], unmanned ground vehicles (UGVs) [7], satellite clusters [8], and so forth. Various consensus control strategies have been proposed and many important results have been achieved in the existing literatures on consensus control of MASs, for instance, consensus control of first-order MASs [9], consensus control of second-order MASs [10], and consensus control of higher-order MASs [11]; asymptotic consensus control of MASs [12], finite-time consensus control of MASs [9], and fixed-time consensus control of MASs [13]; and consensus control of MASs with time-delays [14], consensus control of MASs with input saturation [15], and consensus control of MASs with disturbances [16].

It should be pointed out that disturbances are widely present in MASs, where the effects of disturbances on consensus control of MASs cannot be ignored. The performance of MASs may be decreased by disturbances and especially the stability of MASs can be affected by large disturbances [17], which makes it difficult for MASs to reach consensus. Therefore, how to effectively reduce the influence of disturbances on consensus control of MASs has become an important field of current research. In [18], consensus control problems of MASs with switching topologies and external disturbances were transformed into robust H-infinity control problems, which had the leader-follower H-infinity consensus control of linear MASs achieved. In [16] and [19], a sliding mode control method was employed to eliminate the influence of disturbances on consensus control of MASs, which enabled the consensus control and adaptive tracking problem of MASs to be solved. In [20], an output feedback antidisturbance consensus protocol was designed by using a finite-time observer to observe system disturbances and unmeasurable states. Similarly, an extended state observer based on estimated states of neighbor agents was designed in [21] to eliminate the impact of disturbances on consensus control of MASs, aiming at the leader-following rendezvous problem of UAV swarms with time-varying disturbances. An event-triggered consensus problem of discrete-time MASs with impulsive noise under directed switching topologies was investigated in [22], where the control protocol based on maximum correntropy criterion (MCC) was first proposed for the discrete-time MASs to reach consensus.

Manuscript received 3 January 2024; revised 31 March 2024; accepted 10 May 2024. Date of publication 17 June 2024; date of current version 19 February 2025. Recommended by Technical Editor B. Chu and Senior Editor M. Indri. This work was supported in part by the National Key Research and Development Program of China under Grant 2021ZD0112700, in part by the National Natural Science Foundation of China under Grant 61976175 and Grant 61802036, in part by the National Science Foundation of Sichuan of China under Grant 2022NSFSC0462 and Grant 2023NSFSC1985, and in part by the Transformation Program of Scientific and Technological Achievements of Sichuan of China under Grant 2024ZHC0032 and Grant 2024ZHC0017. (Corresponding authors: Jun Liu; Hongbo Li.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TMECH.2024.3404343>.

Digital Object Identifier 10.1109/TMECH.2024.3404343

It has been demonstrated that the antiinterference methods in the above research are effective to address consensus problems of MASs with disturbances. Among numerous antiinterference methods, Han in [23] and [24] proposed an active disturbance rejection control method, which utilized active estimation to compensate for total disturbances including external disturbances, system model uncertainties, and unmodeled dynamics. In this method, an active disturbance rejection controller including a tracking differentiator (TD), an extended state observer (ESO), and a nonlinear state error feedback law (NLSEF) was designed on the basis of the conventional PID controller [23]. The core idea of active disturbance rejection control (ADRC) is to expand the total disturbances to a new state, estimate it in the ESO, and compensate for it in the NLSEF. The ESO is a key part of ADRC, whose function is to estimate the disturbances in the system and replace the error integral feedback. This avoids the shortcomings in PID such as feedback delay, oscillation, and saturation of control quantity caused by the integral feedback [25]. The ADRC method has the advantages of simple structure, superior control performance, wide application range, excellent antiinterference ability and so on, and has been widely used and deeply studied in [26], [27], and [28]. In [27], a controller based on neural network and ADRC was designed to estimate the unknown total disturbances in the system, which achieved the trajectory tracking control of the unmanned helicopter. An ADRC method was applied to steam pressure control in high-order processes, and the input disturbance rejection and measurement noise rejection capabilities of different controllers were analyzed in [28].

Although ADRC offers advantages, it is primarily used for controlling individual agents. To our knowledge, only a few scholars have studied coordinated control of MASs based on ADRC [29], [30], [31], [32]. In [29] and [31], the third-order linear extended state observer (LESO) was used to estimate total disturbances of each second-order agent, which were then compensated for in the control protocol. Meanwhile, the LESO was extended from a third order to a higher order in [32] and [33], enabling the estimation of total disturbances and unknown system states of higher-order uncertain MASs. However, the literatures mentioned above utilized LESOs, which had slow convergence speed [34]. Therefore, a conventional nonlinear extended state observer (NESO) with faster convergence speed was considered to replace the LESO in [30]. In [30], a conventional NESO with the nonlinear convergence function $fal(s, \mu, \delta)$ was used to estimate the uncertainty model in MASs and track the derivatives of the set virtual control variables by TD, which reduced the calculation complexity and achieved containment control. Nevertheless, Pu et al. [35] suggested that the convergence speed of the conventional NESO would be reduced to some extent due to a design problem of the nonlinear convergence function $fal(s, \mu, \delta)$. It is worth noting that although the above literatures have conducted in-depth research on consensus control of MASs with disturbances, they used the LESO or the conventional NESO with slower convergence speed. Therefore, a novel nonlinear extended state observer (NNESO) with faster convergence speed could be a better alternative.

Motivated by the above discussions, active disturbance rejection consensus control of MASs based on an NNESO is

investigated in this article, where ADRC is used to effectively reduce the impact of unknown total disturbances on consensus control of MASs and enhance the antidisturbance performance of MASs. The innovation of this article mainly focuses on the following three aspects.

- 1) A novel nonlinear convergence function $scf(s, a, b)$ is proposed, which effectively improves the convergence speed compared to the conventional nonlinear convergence function $fal(s, \mu, \delta)$.
- 2) An NNESO based on the novel nonlinear convergence function $scf(s, a, b)$ is proposed, which effectively enhances the tracking speed of the NNESO for the total disturbances caused by system model uncertainties, unmodeled dynamics, and external disturbances.
- 3) A consensus control protocol on the basis of the NNESO is proposed, and the conditions are derived for MASs with disturbances under the control protocol and the NNESO to reach consensus.

The rest of this article is organized as follows. The relevant preliminaries involved in this article are explained in Section II. Section III presents the main problem that this article aims to solve, and designs a control protocol. In Section IV, the main results in this article are included. By numerical simulations and actual experiments, the performance differences of different ESOs are compared and the correctness of the theory is verified in Sections V and VI. Finally, Section VII concludes this article.

II. PRELIMINARIES

For further analysis and research, the knowledge of ADRC theory and algebraic graph theory is introduced in this section.

A. ADRC Theory

A controller that can automatically estimate and compensate for disturbances is called an active disturbance rejection controller, which is mainly composed of an ESO for estimating disturbances and an NLSEF for compensating for disturbances [23].

Consider the following first-order controlled system:

$$\begin{cases} \dot{x}_1(t) = hu(t) + f(x_1(t), t), \\ y(t) = x_1(t) \end{cases} \quad (1)$$

where h is the control coefficient, and $f(x_1(t), t)$ is the total and unknown disturbances. If $f(x_1(t), t)$ is defined as $x_2(t)$, then system (1) can be rewritten as

$$\begin{cases} \dot{x}_1(t) = hu(t) + x_2(t), \\ \dot{x}_2(t) = w_0(t), \\ y(t) = x_1(t) \end{cases} \quad (2)$$

where $w_0(t)$ is the derivative of $f(x_1(t), t)$ with respect to time, and a linear state observer [36] of system (2) or an LESO of system (1) is established as follows:

$$\begin{cases} e_{1-\text{LESO}}(t) = z_1(t) - x_1(t), \\ \dot{z}_1(t) = z_2(t) + hu(t) - \beta_1 e_{1-\text{LESO}}(t), \\ \dot{z}_2(t) = -\beta_2 e_{1-\text{LESO}}(t) \end{cases} \quad (3)$$

where β_1 and β_2 are the coefficients of the observer, $z_1(t)$ and $z_2(t)$ are the states of the observer used to respectively track

$x_1(t)$ and $x_2(t)$ of system (2), and $e_{1-\text{LESO}}(t)$ is the error of $z_1(t)$ tracking $x_1(t)$.

In order to improve the estimation efficiency of LESO (3), $fal(s, \mu, \delta)$ is quoted to obtain conventional NESO (4) [23], which is described as

$$\begin{cases} e_{1-\text{NESO}}(t) = z_1(t) - x_1(t), \\ \dot{z}_1(t) = z_2(t) + hu(t) - \beta_1 fal(e_{1-\text{NESO}}, \mu, \delta), \\ \dot{z}_2(t) = -\beta_2 fal(e_{1-\text{NESO}}, \mu, \delta) \end{cases} \quad (4)$$

where

$$fal(s, \mu, \delta) = \begin{cases} |s|^\mu sgn(s), & |s| > \delta, \\ \frac{s}{\delta^{1-\mu}}, & |s| \leq \delta, \end{cases}$$

$$sgn(s) = \begin{cases} 1, & s > 0, \\ 0, & s = 0, \\ -1, & s < 0 \end{cases}$$

δ is an arbitrarily small positive number, and $\mu \in (0, 1)$.

Although NESO (4) has stronger dynamic performance compared to LESO (3) [34], its convergence speed can be further speeded up. In this article, a novel nonlinear convergence function $scf(s, a, b)$ will be proposed, and it converges faster than $fal(s, \mu, \delta)$ in NESO (4). Based on $scf(s, a, b)$, we will design an NNESO with a faster convergence speed, which can be formulated as

$$\begin{cases} e_{1-\text{NNESO}}(t) = z_1(t) - x_1(t), \\ \dot{z}_1(t) = z_2(t) + hu(t) \\ \quad - \beta_1 scf(e_{1-\text{NNESO}}(t), a, b), \\ \dot{z}_2(t) = -\beta_2 scf(e_{1-\text{NNESO}}(t), a, b). \end{cases} \quad (5)$$

Based on the estimation of the total disturbances by NNESO (5), system (1) can be transformed into a simple error feedback control system.

B. Algebraic Graph Theory

The network topology among agents is represented by graph $\mathcal{G} = \{\mathcal{V}, \varepsilon, \mathcal{A}\}$, where $\mathcal{V} = \{1, 2, \dots, N\}$ is the set of nodes in the graph, $\varepsilon \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges in the graph, and $\mathcal{A} = [a_{ij}]$ is the weighted adjacency matrix of the graph. If there is information communication from node j to i , then $a_{ij} = 1$, otherwise $a_{ij} = 0$. Suppose node i has no self-loops, i.e., $a_{ii} = 0$. $N_i = \{j \in \mathcal{V} | (j, i) \in \varepsilon, j \neq i\}$ represents the index set of neighbors of node i . If $a_{ij} = a_{ji} > 0$, then $\mathcal{G}$ is an undirected graph, and if there exists any $a_{ij} \neq a_{ji}$, then $\mathcal{G}$ is a directed graph. An undirected graph is said to be connected if there is at least one path between any two nodes of the graph. A directed graph has only one spanning tree if at least one node has a directed path to all other nodes.

Define Laplacian matrix $L = D - \mathcal{A}$ of graph $\mathcal{G}$, where $D = \text{diag}\{d_1, d_2, \dots, d_N\}$ is a diagonal matrix with diagonal elements $d_1, d_2, \dots, d_N$, and $d_i = \sum_{j=1}^N a_{ij}$, $i \in \mathcal{V}$. Thus, the element of L is

$$l_{ij} = \begin{cases} \sum_{j=1}^N a_{ij}, & i = j, \\ -a_{ij}, & i \neq j. \end{cases}$$

L can be assigned by L_1 or L_2 , where L_1 represents the Laplacian matrix of an undirected and connected graph, and

L_2 represents the Laplacian matrix of a directed graph with a spanning tree.

Lemma 1 ([37]): For an undirected and connected graph, if L_1 represents the Laplacian matrix of the undirected graph, then 0 is an eigenvalue of L_1 , and there are $L_1 = L_1^T$ and $1_N^T L_1 = 0_N^T$. Moreover, there is a matrix $Q \in \mathbb{R}^{N(N-1)}$ such that $1_N^T Q = 0_{N-1}^T$, $Q^T Q = I_{N-1}$, $Q Q^T = I_N - \frac{1}{N} \times 1_N \times 1_N^T$, and $Q^T L_1 Q$ is a positive definite matrix.

Lemma 2 ([17]): For a directed graph with a spanning tree, if L_2 represents the Laplacian matrix of the directed graph, then L_2 is a positive semi-definite matrix. All eigenvalues other than one 0 have positive real parts, and the eigenvector corresponding to 0 is 1_N .

Lemma 3 ([38]): Consider a system $\dot{x}(t) = Ax(t) + Cr(t)$, if the matrix A is Hurwitz stable, then there exists the following inequality:

$$\|x(t)\| \leq ce^{-\epsilon t} \|x(0)\| + \|C\| \left(\int_0^\infty \|e^{A\tau}\| d\tau \right) \|r(t)\|_\infty \quad (6)$$

where c and ϵ are given positive numbers. If $r(t) \equiv 0$, then the necessary and sufficient condition for the system to achieve stability is that A is Hurwitz stable.

Notation: In this article, $\mathbb{R}^n$ represents an n -dimensional real number column vector, and $\mathbb{R}^{N \times n}$ represents an $N \times n$ -dimensional real number matrix. N represents the number of agents in MASs. 1_N refers to an N -dimensional column vector whose elements are all 1. I_n refers to an n -dimensional identity matrix. $R(P)$ represents the rank of matrix P . $\|\cdot\|$ represents the 2-norm of a vector or matrix. $\|\cdot\|_\infty$ represents the infinite norm of a vector or matrix, and $\otimes$ is Kronecker product.

III. PROBLEM STATEMENT

Consider MASs with total disturbances, which consist of N agents. The network topology between agents can be represented by graph $\mathcal{G}$. The dynamic model for each agent can be described as

$$\begin{cases} \dot{x}_i(t) = u_i(t) + d_i(t), \\ y_i(t) = x_i(t) \end{cases} \quad (7)$$

where the subscript $i \in \mathcal{V}$ represents the i th agent, $x_i(t) \in \mathbb{R}^n$ is the state of the i th agent, $u_i(t) \in \mathbb{R}^n$ is the control input, and $d_i(t) \in \mathbb{R}^n$ are the unknown total disturbances caused by system model uncertainties, unmodeled dynamics and external disturbances.

Before a consensus control protocol of MASs (7) is introduced, the following definition of consensus is first given.

Definition 1 ([39]): The consensus definition of MASs (7) is that under any bounded initial condition, MASs (7) need to satisfy

$$\lim_{t \rightarrow +\infty} \|x_i(t) - x_j(t)\| \leq \rho, \quad \forall i, j \in \mathcal{V} \quad (8)$$

where ρ is a nonnegative constant in an arbitrarily assigned neighborhood, and represents the maximum allowable state error of the MASs.

Since there are unknown disturbances in MASs (7), the following control protocol with disturbance compensation is given

as

$$u_i(t) = -k \sum_{j \in N_i} a_{ij} (x_i(t) - x_j(t)) - z_{i2}(t) \quad (9)$$

where $k > 0$ is the control gain, $-k \sum_{j \in N_i} a_{ij} (x_i(t) - x_j(t))$ is the consensus term that makes all states of MASs (7) converge to a same value, and $z_{i2}(t)$ is an estimator of $d_i(t)$ which can be determined by referring to (5).

The $scf(s, a, b)$ in (5) is designed as

$$scf(s, a, b) = \eta \left(|s|^a + a^{\frac{a}{|a|+b}} |s|^b \right) \text{sgn}(s) \quad (10)$$

where $\eta \geq 1$, $a > 1$ and $b \in (0, 1)$. By comparing $fal(s, \mu, \delta)$ in NESO (4) and $scf(s, a, b)$, it can be obtained that the convergence speed of $scf(s, a, b)$ is faster than that of $fal(s, \mu, \delta)$ under $b = \mu$, which satisfies the design requirements. The relevant proof is shown in the Appendix. When $s \in \mathbb{R}^n$, one can define $scf(s, a, b) = [scf(s_1, a, b), scf(s_2, a, b), \dots, scf(s_n, a, b)]^T$. The NNESO based on $scf(s, a, b)$ for MASs (7) is designed as

$$\begin{cases} e_{i1-\text{NNESO}}(t) = z_{i1}(t) - x_i(t), \\ \dot{z}_{i1}(t) = z_{i2}(t) + u_i(t) \\ \quad - \beta_{i1} scf(e_{i1-\text{NNESO}}(t), a, b), \\ \dot{z}_{i2}(t) = -\beta_{i2} scf(e_{i1-\text{NNESO}}(t), a, b) \end{cases} \quad (11)$$

where $\beta_{i1}, \beta_{i2} > 0$ are the observer coefficients of the i th agent, $z_{i1}(t) \in \mathbb{R}^n$ and $z_{i2}(t) \in \mathbb{R}^n$ are the states of the observer used to track $x_i(t)$ and $d_i(t)$, respectively, and $e_{i1-\text{NNESO}}(t)$ is the error of $z_{i1}(t)$ tracking $x_i(t)$.

After $z_{i2}(t)$ is obtained according to NNESO (11), the following closed-loop model of MASs (7) under the control protocol (9) can be formulated as

$$\dot{x}_i(t) = -k \sum_{j \in N_i} a_{ij} (x_i(t) - x_j(t)) + d_i(t) - z_{i2}(t). \quad (12)$$

Let $x(t) = [x_1^T(t), x_2^T(t), \dots, x_N^T(t)]^T$, $e_{2-\text{NNESO}}(t) = [e_{12-\text{NNESO}}^T(t), e_{22-\text{NNESO}}^T(t), \dots, e_{N2-\text{NNESO}}^T(t)]^T$, where $x(t) \in \mathbb{R}^{N \times n}$, $e_{2-\text{NNESO}}(t) \in \mathbb{R}^{N \times n}$, and $e_{i2-\text{NNESO}}(t) = z_{i2}(t) - d_i(t)$. From the above definition, the closed-loop model of MASs (12) can be written in the following compact form:

$$\dot{x}(t) = -k(L \otimes I_n)x(t) - e_{2-\text{NNESO}}(t). \quad (13)$$

Define

$$\bar{x}_i(t) = x_i(t) - \frac{1}{N} \sum_{j=1}^N x_j(t)$$

and the above formula is converted as

$$\bar{x}(t) = (P \otimes I_n)x(t) \quad (14)$$

where $\bar{x}(t) = [\bar{x}_1^T(t), \bar{x}_2^T(t), \dots, \bar{x}_N^T(t)]^T$, $P = I_N - \frac{1}{N} \times 1_N \times 1_N^T$, $P = P^T$, $LP = L$, and $R(P) = N - 1$.

The derivative of $\bar{x}(t)$ in (14) is

$$\begin{aligned} \dot{\bar{x}}(t) &= -(P \otimes I_n)(k(L \otimes I_n)x(t) + e_{2-\text{NNESO}}(t)) \\ &= -k(PLP \otimes I_n)x(t) - (P \otimes I_n)e_{2-\text{NNESO}}(t) \\ &= -k(PL \otimes I_n)\bar{x}(t) - (P \otimes I_n)e_{2-\text{NNESO}}(t). \end{aligned} \quad (15)$$

Remark 1: It can be seen that there is an estimator $z_{i2}(t)$ for the total disturbances $d_i(t)$ in the control protocol $u_i(t)$. The influence of $d_i(t)$ on consensus control of MASs (7) can be eliminated when the error of $z_{i2}(t)$ tracking $d_i(t)$ is enough small. $z_{i2}(t)$ can be calculated according to $x_i(t)$ obtained from a sensor feedback by NNESO (11).

IV. MAIN RESULTS

In this section, the conditions are derived for MASs (7) under control protocol (9) and NNESO (11) to reach consensus in the case of an undirected and connected topology or a directed topology with a spanning tree when the derivatives of the total disturbances are equal to zero or not.

A. Consensus of MASs Over an Undirected and Connected Topology

Theorem 1: Consider MASs (7) whose network topology obeys Lemma 1. If $\dot{d}_i(t) = 0$, then MASs (7) under control protocol (9) and NNESO (11) can reach consensus.

Proof: According to Lemma 1, substituting $PL_1 = L_1$ and $P = QQ^T$ into (15) can be rewritten as

$$\dot{\bar{x}}(t) = -k(L_1 \otimes I_n)\bar{x}(t) - (QQ^T \otimes I_n)e_{2-\text{NNESO}}(t). \quad (16)$$

Define $\tilde{x}(t) = (Q^T \otimes I_n)\bar{x}(t)$, and take the time derivative on both sides of the equation, it can be obtained that

$$\dot{\tilde{x}}(t) = -k(Q^T L_1 \otimes I_n)\tilde{x}(t) - (Q^T \otimes I_n)e_{2-\text{NNESO}}(t). \quad (17)$$

Since

$$\begin{aligned} (Q^T L_1 Q \otimes I_n)\tilde{x}(t) &= (Q^T L_1 Q \otimes I_n)(Q^T \otimes I_n)\bar{x} \\ &= (Q^T L_1 \otimes I_n)\bar{x} \end{aligned} \quad (18)$$

equation (17) can be rewritten as

$$\begin{aligned} \dot{\tilde{x}}(t) &= -k(Q^T L_1 Q \otimes I_n)\tilde{x}(t) \\ &\quad - (Q^T \otimes I_n)e_{2-\text{NNESO}}(t). \end{aligned} \quad (19)$$

If the following Lyapunov candidate function is designed as

$$V_1(t) = \tilde{x}(t)^T \tilde{x}(t) \quad (20)$$

then it is easy to get

$$\begin{aligned} \dot{V}_1(t) &= \dot{\tilde{x}}^T(t)\tilde{x}(t) + \tilde{x}^T(t)\dot{\tilde{x}}(t) \\ &= -2k\tilde{x}^T(t)(Q^T L_1 Q \otimes I_n)\tilde{x}(t) \\ &\quad - e_{2-\text{NNESO}}^T(t)(Q \otimes I_n)\tilde{x}(t) \\ &\quad - \tilde{x}^T(t)(Q^T \otimes I_n)e_{2-\text{NNESO}}(t) \end{aligned} \quad (21)$$

where

$$e_{2-\text{NNESO}}^T(t)(Q \otimes I_n)\tilde{x}(t) \leq \|e_{2-\text{NNESO}}(t)\| \|\tilde{x}(t)\| \quad (22)$$

and

$$\tilde{x}^T(t)(Q^T \otimes I_n)e_{2-\text{NNESO}}(t) \leq \|e_{2-\text{NNESO}}(t)\| \|\tilde{x}(t)\| \quad (23)$$

according to $\|Q^T \otimes I_n\| = 1$ and the properties of the matrix norm.

Combining (21), (22), and (23), it is easy to obtain

$$\begin{aligned} \dot{V}_1(t) \leq & -2k\tilde{x}^T(t)(Q^T L_1 Q \otimes I_n)\tilde{x}(t) \\ & + 2\|e_{2-\text{NNESO}}(t)\|\|\tilde{x}(t)\|. \end{aligned} \quad (24)$$

According to MASs (7) and NNESO (11), one can get

$$\begin{aligned} \dot{e}_{i1-\text{NNESO}}(t) &= \dot{z}_{i1}(t) - \dot{x}_i(t) \\ &= z_{i2}(t) - d_i(t) \\ &\quad - \beta_{i1} \text{scf}(e_{i1-\text{NNESO}}(t), a, b) \end{aligned} \quad (25)$$

and

$$\begin{aligned} \dot{e}_{i2-\text{NNESO}}(t) &= \dot{z}_{i2}(t) - \dot{d}_i(t) \\ &= -\beta_{i2} \text{scf}(e_{i1-\text{NNESO}}(t), a, b) - \dot{d}_i(t). \end{aligned} \quad (26)$$

When NNESO (11) reaches steady states, $\dot{e}_{i1-\text{NNESO}}(t) = 0$ and $\dot{e}_{i2-\text{NNESO}}(t) = 0$. Since $\dot{d}_i(t) = 0$, there are the formulas

$$\lim_{t \rightarrow +\infty} \|\text{scf}(e_{i1-\text{NNESO}}(t), a, b)\| = 0$$

and

$$\lim_{t \rightarrow +\infty} \|z_{i2}(t) - d_i(t)\| = 0$$

which lead to

$$\lim_{t \rightarrow +\infty} \|e_{2-\text{NNESO}}(t)\| = \lim_{t \rightarrow +\infty} \|z_2(t) - d(t)\| = 0. \quad (27)$$

When NNESO (11) reaches steady states, according to (24), (27), Lemma 1, and $-2k < 0$, one can obtain

$$\dot{V}_1(t) \leq -2k\tilde{x}^T(t)(Q^T L_1 Q \otimes I_n)\tilde{x}(t) < 0$$

which indicates that MASs (7) whose network topology obeys Lemma 1 can reach consensus without state errors under constant disturbances. The proof is completed.

Theorem 2: Consider MASs (7) whose network topology obeys Lemma 1. If $\sup\{\|\dot{d}_i(t) \neq 0\|, \forall t\} \leq \varepsilon_i (\exists \varepsilon_i > 0)$, then

$$\lim_{t \rightarrow +\infty} \|x_i(t) - x_j(t)\| \leq \frac{\sqrt{2}}{k\|L_1\|} \left(\sum_{i=1}^N \left(\frac{\beta_{i1}}{\beta_{i2}} \right)^2 \varepsilon_i^2 \right)^{\frac{1}{2}} \quad (28)$$

which indicates that MASs (7) under control protocol (9) and NNESO (11) can reach consensus.

Proof: When NNESO (11) reaches steady states, (27) holds under the condition of Theorem 1, which makes that the nonlinear term $e_{2-\text{NNESO}}(t)$ can be eliminated, and system (16) can be simplified as the following linear system

$$\dot{\tilde{x}}(t) = -k(L_1 \otimes I_n)\tilde{x}(t). \quad (29)$$

It can be concluded from Theorem 1, Lemma 3, and (29) that $-k(L_1 \otimes I_n)$ is Hurwitz stable, which indicates that system (15) satisfies the condition of Lemma 3. According to (6) and system (15), one can get

$$\begin{aligned} \|\tilde{x}(t)\| &\leq ce^{-\epsilon t}\|\tilde{x}(0)\| \\ &\quad + \|QQ^T \otimes I_n\| \left(\int_0^\infty \|e^{-k(L_1 \otimes I_n)\tau}\| d\tau \right) \\ &\quad \|e_{2-\text{NNESO}}(t)\|_\infty \end{aligned} \quad (30)$$

and

$$\begin{aligned} \lim_{t \rightarrow +\infty} \|\tilde{x}(t)\| &\leq \lim_{t \rightarrow +\infty} ce^{-\epsilon t}\|\tilde{x}(0)\| \\ &\quad + \lim_{t \rightarrow +\infty} \|QQ^T\| \left(\int_0^\infty \|e^{-kL_1\tau}\| d\tau \right) \\ &\quad \|e_{2-\text{NNESO}}(t)\|_\infty. \end{aligned} \quad (31)$$

Since $\lim_{t \rightarrow +\infty} ce^{-\epsilon t} = 0$, $\lim_{\tau \rightarrow +\infty} e^{-kL_1\tau} = 0$, $\lim_{\tau \rightarrow 0} e^{-kL_1\tau} = 1$ and $\|QQ^T\| = 1$, (31) can be simplified as

$$\begin{aligned} \lim_{t \rightarrow +\infty} \|\tilde{x}(t)\| &\leq \frac{1}{k\|L_1\|} \lim_{t \rightarrow +\infty} \|e_{2-\text{NNESO}}(t)\|_\infty \\ &\leq \frac{1}{k\|L_1\|} \lim_{t \rightarrow +\infty} \|e_{2-\text{NNESO}}(t)\|. \end{aligned} \quad (32)$$

When NNESO (11) reaches steady states, (25) and (26) can be transformed into the following formulas:

$$\lim_{t \rightarrow +\infty} (-\beta_{i2} \text{scf}(e_{i1-\text{NNESO}}(t), a, b) - \dot{d}_i(t)) = 0,$$

$$\lim_{t \rightarrow +\infty} \|\beta_{i2} \text{scf}(e_{i1-\text{NNESO}}(t), a, b)\| = \lim_{t \rightarrow +\infty} \|\dot{d}_i(t)\| \leq \varepsilon_i,$$

$$\lim_{t \rightarrow +\infty} (z_{i2}(t) - d_i(t) - \beta_{i1} \text{scf}(e_{i1-\text{NNESO}}(t), a, b)) = 0$$

and

$$\lim_{t \rightarrow +\infty} \|z_{i2}(t) - d_i(t)\| \leq \frac{\beta_{i1}}{\beta_{i2}} \varepsilon_i$$

which lead to

$$\begin{aligned} \lim_{t \rightarrow +\infty} \|e_{2-\text{NNESO}}(t)\| &= \lim_{t \rightarrow +\infty} \|z_2(t) - d(t)\| \\ &\leq \left(\sum_{i=1}^N \left(\frac{\beta_{i1}}{\beta_{i2}} \right)^2 \varepsilon_i^2 \right)^{\frac{1}{2}}. \end{aligned} \quad (33)$$

Since there are $\sqrt{2}\|\tilde{x}(t)\| \geq \sqrt{2\tilde{x}_i^T(t)\tilde{x}_i(t) + 2\tilde{x}_j^T(t)\tilde{x}_j(t)}$
 $\geq \sqrt{\tilde{x}_i^T(t)\tilde{x}_i(t) + \tilde{x}_j^T(t)\tilde{x}_j(t) - 2\tilde{x}_i^T(t)\tilde{x}_j(t)} = \|x_i(t) - x_j(t)\|$, (32), and (33), one can get

$$\begin{aligned} \lim_{t \rightarrow +\infty} \|x_i(t) - x_j(t)\| &\leq \sqrt{2} \lim_{t \rightarrow +\infty} \|\tilde{x}(t)\| \\ &\leq \frac{\sqrt{2}}{k\|L_1\|} \left(\sum_{i=1}^N \left(\frac{\beta_{i1}}{\beta_{i2}} \right)^2 \varepsilon_i^2 \right)^{\frac{1}{2}} \end{aligned} \quad (34)$$

which indicates that MASs (7) whose network topology obeys Lemma 1 can reach consensus under the existence of derivatives of total disturbances with upper bounds. The proof is completed.

Remark 2: From the above proof, it is easy for us to derive that the existence of the derivative of total disturbances $d_i(t)$ are necessary. When the derivative does not exist, $\|z_{i2}(t) - d_i(t)\| \rightarrow \infty$, i.e., NNESO (11) cannot track the total disturbances $d_i(t)$. If the derivative exists, then the state errors can be made enough small by choosing appropriate parameters. This makes MASs (7) satisfy the definition of consensus. It is worth noting that the time for MASs (7) to achieve consensus is related to the speed of NNESO (11) tracking the total disturbances. Only when NNESO (11) reaches stable states, (27) and (34) are established, and then MASs (7) can reach consensus.

B. Consensus of MASs Over a Directed Topology With a Spanning Tree

Theorem 3: Consider MASs (7) whose network topology obeys Lemma 2. If $\dot{d}_i(t) = 0$, then MASs (7) under control protocol (9) and NNESO (11) can reach consensus.

Proof: Let $P = \begin{bmatrix} \frac{N-1}{N} & p^r \\ p^c & P^{cr} \end{bmatrix}$ and $L_2 = \begin{bmatrix} 0 & 0_{1 \times (N-1)} \\ l_2^c & L_2^{cr} \end{bmatrix}$, where $p^r \in \mathbb{R}^{1 \times (N-1)}$, $p^c \in \mathbb{R}^{(N-1) \times 1}$, $l_2^c \in \mathbb{R}^{(N-1) \times 1}$, $P^{cr} \in \mathbb{R}^{(N-1) \times (N-1)}$, and $L_2^{cr} \in \mathbb{R}^{(N-1) \times (N-1)}$. From Lemma 2, we can obtain that both P^{cr} and L_2^{cr} are full rank matrices, thus, one has

$$\begin{aligned} \min(R(P), R(L_2)) &\geq R(PL_2) = R\left(\begin{bmatrix} p^r l_2^c & p^r L_2^{cr} \\ P^{cr} l_2^c & P^{cr} L_2^{cr} \end{bmatrix}\right) \\ &\geq R(P^{cr} L_2^{cr}) = N - 1 \end{aligned} \quad (35)$$

which indicates that $R(PL_2) = N - 1$. It can be assumed that $\lambda_i (i \in \{2, 3, \dots, N\})$ is a nonzero eigenvalue of PL_2 . There is a matrix $U \in \mathbb{R}^{N \times N}$, so that PL_2 is transformed into the following form:

$$U^{-1}PL_2U = J = \begin{bmatrix} 0_{1 \times 1} & 0_{1 \times (N-1)} \\ 0_{(N-1) \times 1} & \Delta \end{bmatrix} \quad (36)$$

where J is a semipositive definite matrix and Δ is the Jordan standard form composed of $\lambda_2, \lambda_3, \dots, \lambda_N$.

Introducing the state transformation $\hat{x}(t) = (U^{-1} \otimes I_n)\bar{x}(t)$, (15) can be rewritten as

$$\begin{aligned} \dot{\hat{x}}(t) &= -k(U^{-1}PL_2U \otimes I_n)(U^{-1} \otimes I_n)\bar{x}(t) \\ &\quad - (U^{-1}P \otimes I_n)e_{2-\text{NNESO}}(t) \\ &= -k(J \otimes I_n)\hat{x}(t) - (U^{-1}P \otimes I_n)e_{2-\text{NNESO}}(t). \end{aligned} \quad (37)$$

If the following Lyapunov candidate function is designed as

$$V_2(t) = \hat{x}(t)^T \hat{x}(t) \quad (38)$$

then it is easy to get

$$\begin{aligned} \dot{V}_2(t) &= \dot{\hat{x}}^T(t)\hat{x}(t) + \hat{x}^T(t)\dot{\hat{x}}(t) \\ &= -2k\hat{x}^T(t)(J \otimes I_n)\hat{x}(t) \\ &\quad - e_{2-\text{NNESO}}^T(t)((U^{-1}P)^T \otimes I_n)\hat{x}(t) \\ &\quad - \hat{x}^T(t)(U^{-1}P \otimes I_n)e_{2-\text{NNESO}}(t) \end{aligned} \quad (39)$$

where

$$e_{2-\text{NNESO}}^T(t)((U^{-1}P)^T \otimes I_n)\hat{x}(t) \leq \|\hat{x}(t)\| \|U^{-1}P\| \|e_{2-\text{NNESO}}(t)\| \quad (40)$$

and

$$\hat{x}^T(t)(U^{-1}P \otimes I_n)e_{2-\text{NNESO}}(t) \leq \|\hat{x}(t)\| \|U^{-1}P\| \|e_{2-\text{NNESO}}(t)\|. \quad (41)$$

When NNESO (11) reaches steady states, according to (27), (39), (40), and (41), one can obtain

$$\begin{aligned} \dot{V}_2(t) &\leq -2k\hat{x}^T(t)(J \otimes I_n)\hat{x}(t) \\ &\quad + 2\|e_{2-\text{NNESO}}(t)\| \|\hat{x}(t)\| \|U^{-1}P\| \end{aligned}$$

$$= -2k\hat{x}^T(t)(J \otimes I_n)\hat{x}(t). \quad (42)$$

Since $k > 0$, J is a semipositive definite matrix, and $\dot{V}_2(t) \equiv 0$ iff $\hat{x}(t) \equiv 0_N$, then $\dot{V}_2(t) \leq 0$, which indicates that MASs (7) whose network topology obeys Lemma 2 can reach consensus without state errors under constant disturbances. The proof is completed.

Theorem 4: Consider MASs (7) whose network topology obeys Lemma 2. If $\sup\{\|\dot{d}_i(t) \neq 0\|, \forall t\} \leq \varepsilon_i (\exists \varepsilon_i > 0)$, then

$$\lim_{t \rightarrow +\infty} \|x_i(t) - x_j(t)\| \leq \frac{\sqrt{2}}{k\|PL_2\|} \left(\sum_{i=1}^N \left(\frac{\beta_{i1}}{\beta_{i2}} \right)^2 \varepsilon_i^2 \right)^{\frac{1}{2}} \quad (43)$$

which indicates that MASs (7) under control protocol (9) and NNESO (11) can reach consensus.

Proof: Similar to Theorem 2, when NNESO (11) reaches stable states and MASs (7) satisfy the conditions in Theorem 3, system (15) can be simplified as the following linear system

$$\dot{\bar{x}}(t) = -k(PL_2 \otimes I_n)\bar{x}(t). \quad (44)$$

It can be concluded from Theorem 3, Lemma 3, and (44) that $-k(PL_2 \otimes I_n)$ is Hurwitz stable, which indicates that system (15) satisfies the condition of Lemma 3. According to $\|P \otimes I_n\| = 1$, (6), and (15), one can get

$$\begin{aligned} \|\bar{x}(t)\| &\leq ce^{-\epsilon t} \|\bar{x}(0)\| \\ &\quad + \|P \otimes I_n\| \left(\int_0^\infty \|e^{-k(PL_2 \otimes I_n)\tau}\| d\tau \right) \\ &\quad \|e_{2-\text{NNESO}}(t)\|_\infty \end{aligned} \quad (45)$$

and

$$\begin{aligned} \lim_{t \rightarrow +\infty} \|\bar{x}(t)\| &\leq \lim_{t \rightarrow +\infty} ce^{-\epsilon t} \|\bar{x}(0)\| \\ &\quad + \lim_{t \rightarrow +\infty} \left(\int_0^\infty \|e^{-kPL_2\tau}\| d\tau \right) \\ &\quad \|e_{2-\text{NNESO}}(t)\|_\infty \\ &\leq \frac{1}{k\|PL_2\|} \lim_{t \rightarrow +\infty} \|e_{2-\text{NNESO}}(t)\|. \end{aligned} \quad (46)$$

Since there exists the derivative of the total disturbances that has an upper bound, (33) holds. From (32) to (34), we can get that the formula for calculating the maximum state error of MASs (7) over a directed topology with a spanning tree is similar as that of MASs (7) over an undirected and connected graph, which indicates that it is easy to derive $\|x_i(t) - x_j(t)\| \leq \frac{\sqrt{2}}{k\|PL_2\|} \left(\sum_{i=1}^N \left(\frac{\beta_{i1}}{\beta_{i2}} \right)^2 \varepsilon_i^2 \right)^{\frac{1}{2}}$, and MASs (7) whose network topology obeys Lemma 2 can reach consensus under the existence of derivatives of total disturbances with upper bounds. The proof is completed.

V. SIMULATION RESULTS

In this section, the correctness of theorems is verified by numerical simulations containing 12 examples. Suppose that MASs (7) consist of 3 agents. The consensus will be validated

TABLE I
SIMULATION EXPERIMENTS AND RESULTS FOR MASs (7) UNDER AN UNDIRECTED AND CONNECTED NETWORK TOPOLOGY

Example number	Type of observer	Type of disturbances	Observer parameters	Maximum state error	Error ratio/%	Convergence time/s	Time ratio/%	Consensus
1	None	Nonconstant						No
2	LESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 2$	0.00818	100	2.388	100	Yes
3	NESO	Nonconstant	$\delta = 0.1, \mu = 0.5, \beta_{i1} = 2, \beta_{i2} = 2$	0.00630	77.1	2.739	114.7	Yes
4	NNESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 2, a = 2, b = 0.5, \eta = 1$	0.00578	70.7	1.578	66.1	Yes
5	NNESO	Constant	$\beta_{i1} = 2, \beta_{i2} = 2, a = 2, b = 0.5, \eta = 1$	0.0001	0.1	1.584	66.4	Yes
6	NNESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 4, a = 2, b = 0.5, \eta = 1$	0.00363	44.4	1.366	57.3	Yes

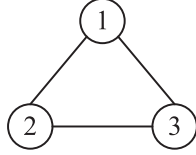


Fig. 1. Undirected and connected network topology.

for MASs (7) without and with disturbance compensation based on LESO (3), NESO (4), and NNESO (11), respectively.

The initial states of agents are set as

$$[x_1(0), x_2(0), x_3(0)] = [5, 3, -5]. \quad (47)$$

The values of the total disturbances are constant in *Example 5* and *11*, and are set as

$$[d_1, d_2, d_3] = [2.1, 1.2, -2.3] \quad (48)$$

while the values of the total disturbances in the remaining examples are nonconstant, and are set as

$$\begin{aligned} d_1(t) &= 0.2 \sin(t + 0.3) + 2.1, \\ d_2(t) &= 0.18 \sin(t + 0.5) + 1.2, \\ d_3(t) &= 0.1 \sin(t - 0.2) - 2.3. \end{aligned} \quad (49)$$

In addition, $n = 1$ and $k = 5$ in all examples. If the maximum distance between the state curves of MASs (7) during convergence is equal to 0.1 at time t_c and the distance between the state curves is less than 0.1 after t_c , then t_c is called the convergence time of consensus of MASs (7). When the convergence time is smaller, the convergence speed of the state of MASs (7) is faster. The state errors in all examples are the steady-state errors of MASs (7).

A. Simulations for Theorems 1 and 2

Consider MASs (7), whose network topology is an undirected and connected graph $\mathcal{G}$ shown in Fig. 1. The Laplacian matrix of graph $\mathcal{G}$ is given as

$$L_1 = \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

Six examples are set up to compare the effects of different observers and different parameters on consensus of MASs (7) under an undirected and connected network topology, and their simulation results are shown in Table I and Fig. 2.

MASs (7) in *Example 1* can not reach consensus due to the lack of estimation and compensation of the total disturbances.

In *Example 2, 3, and 4*, it is obvious that MASs (7) converge to a finite range within the maximum allowable state error, which shows that MASs (7) can achieve consensus. At this time, the maximum state errors are 0.00818, 0.0063, and 0.00578, and the convergence time is 2.388, 2.739, and 1.578 s, respectively. Evidently, MASs (7) with NNESO (11) in *Example 4* have the shortest convergence time and the smallest maximum state error. Therefore, the convergence speed of $scf(s, a, b)$ in NNESO (11) is faster than that of $fal(s, \mu, \delta)$ in conventional NESO (4), which indicates that the conclusion in the Appendix is correct when the network topology of MASs (7) is an undirected and connected graph.

It is obvious from *Example 5* that MASs (7) can achieve consensus without state errors, which indicates that the correctness of Theorem 1 is verified. Since the maximum state error for MASs (7) in steady state is reduced from 0.00578 to 0.00363 and its convergence time is reduced from 1.578 to 1.366 s by reducing the ratio of β_{i1} and β_{i2} , the consensus performance of MASs (7) is improved from *Example 4* to *Example 6*. Although the reduction of the ratio of parameters can reduce the state error and convergence time, it is at the cost of increasing the overshoot, which puts forward higher requirements on the hardware performance of agents. To further verify the correctness of Theorem 2, it is necessary to calculate the theoretical maximum state error by (28). Suppose that the derivative of total disturbances $d_i(t_0)$ of each agent can reach the maximum value ε_i at the same time t_0 . Bringing (49), $k, \|L_1\| = 3, \beta_{i1}$, and β_{i2} into (28), one can get

$$\|\bar{x}_{e1}(t_0)\| \approx 0.0271, \quad (50a)$$

$$\|\bar{x}_{e2}(t_0)\| \approx 0.0135 \quad (50b)$$

where $\|\bar{x}_{e1}(t_0)\|$ and $\|\bar{x}_{e2}(t_0)\|$ are the theoretical maximum state errors under the conditions of *Example 4* and *Example 6*, respectively. Since the maximum state error 0.00578 is less than the theoretical maximum state error 0.0271 in *Example 4* and the maximum state error 0.00363 is less than the theoretical maximum state error 0.0135 in *Example 6*, then the correctness of Theorem 2 is verified.

B. Simulations for Theorems 3 and 4

Consider MASs (7), whose network topology is a directed graph $\mathcal{G}$ with a spanning tree shown in Fig. 3. The Laplacian matrix of graph $\mathcal{G}$ is given as

$$L_2 = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}.$$

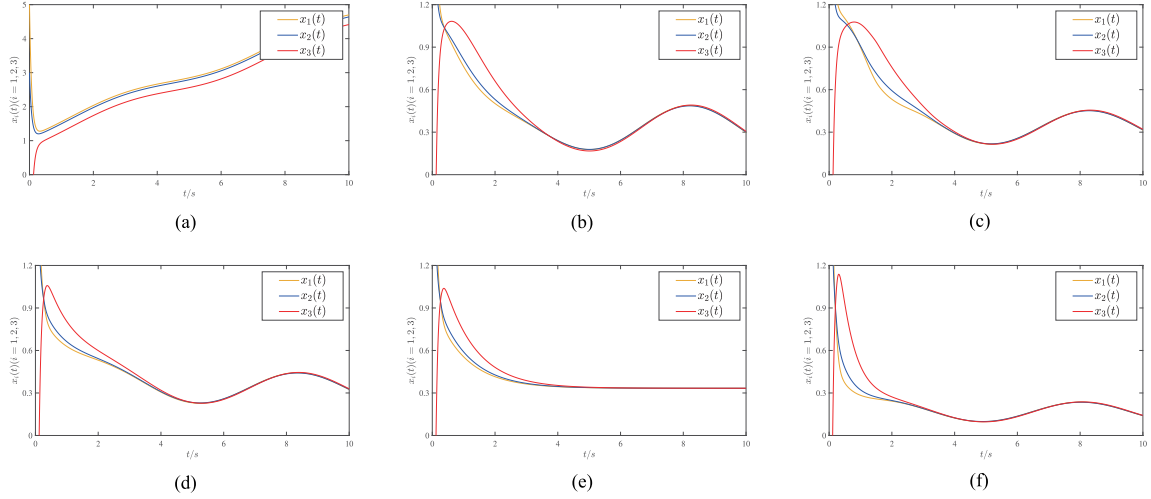


Fig. 2. States trajectory of MASs (7). (a) Example 1. (b) Example 2. (c) Example 3. (d) Example 4. (e) Example 5. (f) Example 6.

TABLE II
SIMULATION EXPERIMENTS AND RESULTS FOR MASs (7) UNDER A DIRECTED NETWORK TOPOLOGY WITH A SPANNING TREE

Example number	Type of observer	Type of disturbances	Observer parameters	Maximum state error	Error ratio/%	Convergence time/s	Time ratio/%	Consensus
7	None	Nonconstant						No
8	LESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 2$	0.01905	100	2.987	100	Yes
9	NESO	Nonconstant	$\delta = 0.1, \mu = 0.5, \beta_{i1} = 2, \beta_{i2} = 2$	0.01277	67.1	3.326	111.4	Yes
10	NNESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 2, a = 2, b = 0.5, \eta = 1$	0.01171	61.5	2.565	85.9	Yes
11	NNESO	Constant	$\beta_{i1} = 2, \beta_{i2} = 2, a = 2, b = 0.5, \eta = 1$	0.0001	0.1	2.495	83.6	Yes
12	NNESO	Nonconstant	$\beta_{i1} = 2, \beta_{i2} = 4, a = 2, b = 0.5, \eta = 1$	0.00737	38.7	1.918	64.3	Yes

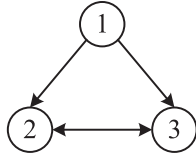


Fig. 3. Directed network topology with a spanning tree.

Six examples are set up to compare the effects of different observers and different parameters on consensus of MASs (7) under a directed network topology with a spanning tree, and their simulation results are shown in Table II and Fig. 4.

MASs (7) in Example 7 are also unable to reach consensus due to the lack of estimation and compensation of the total disturbances. In Example 8, 9, and 10, it is obvious that MASs (7) converge to a finite range within the maximum allowable state error, which shows that MASs (7) can achieve consensus. At this time, the maximum state errors are 0.01905, 0.01277, and 0.01171, and the convergence time is 2.987, 3.326, and 2.565 s, respectively. Evidently, MASs (7) with NNESO (11) in Example 10 have the shortest convergence time and the smallest maximum state error. Therefore, the convergence speed of $scf(s, a, b)$ in NNESO (11) is faster than that of $fal(s, \mu, \delta)$ in conventional NESO (4), which indicates that the conclusion in the Appendix is correct when the network topology of MASs (7) is a directed graph with a spanning tree.

It is obvious from Example 11 that MASs (7) can still achieve consensus without state errors, which indicates that the correctness of Theorem 3 is verified. The maximum state error of MASs (7) is reduced from 0.01171 to 0.00737 and its convergence time is reduced from 2.565 to 1.918 s by reducing the ratio of β_{i1} and β_{i2} . Therefore, the consensus performance of MASs (7) is improved from Example 10 to Example 12. To further verify the correctness of Theorem 4, it is necessary to calculate the theoretical maximum state error by (43). Suppose that the derivative of total disturbances $d_i(t_0)$ of each agent can reach the maximum value ε_i at the same time t_0 . Bringing (49), k , $\|PL_2\| = 3$, β_{i1} , and β_{i2} into (43), one can get

$$\|\bar{x}_{e3}(t_0)\| \approx 0.0271, \quad (51a)$$

$$\|\bar{x}_{e4}(t_0)\| \approx 0.0135 \quad (51b)$$

where $\|\bar{x}_{e3}(t_0)\|$ and $\|\bar{x}_{e4}(t_0)\|$ are the theoretical maximum state errors under the conditions of Example 10 and Example 12, respectively. Since the maximum state error 0.01171 is less than the theoretical maximum state error 0.0271 in Example 10 and the maximum state error 0.00737 is less than the theoretical maximum state error 0.0135 in Example 12, then the correctness of Theorem 4 is verified.

VI. EXPERIMENT DEMONSTRATION

In the experiments, there are three self-made wheeled robots which are used as experimental objects, where each robot is

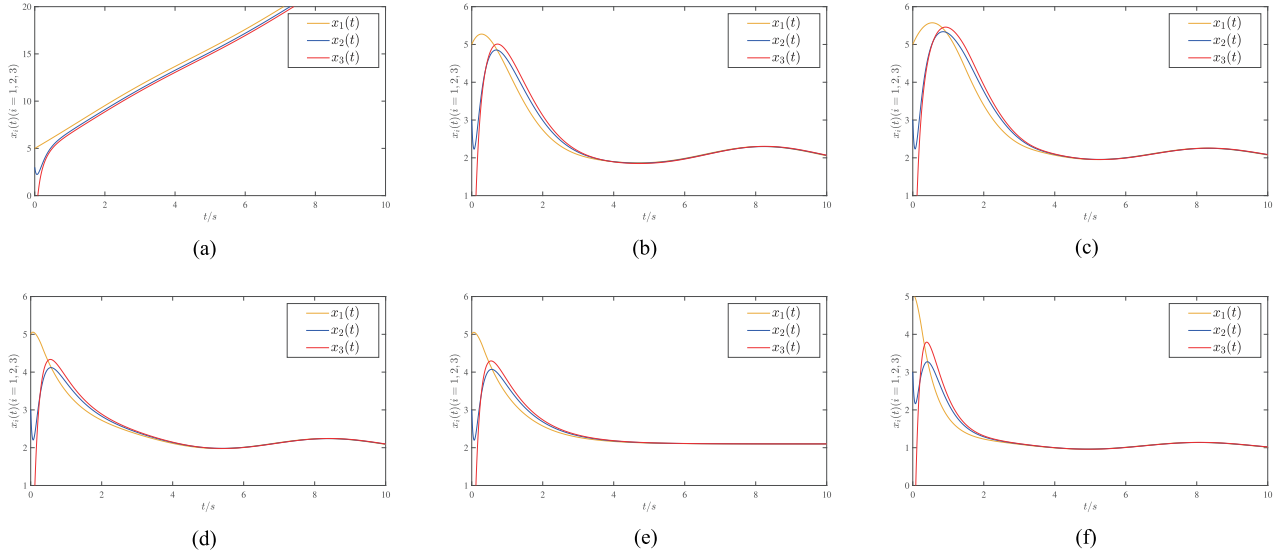


Fig. 4. States trajectory of MASs (7). (a) Example 7. (b) Example 8. (c) Example 9. (d) Example 10. (e) Example 11. (f) Example 12.

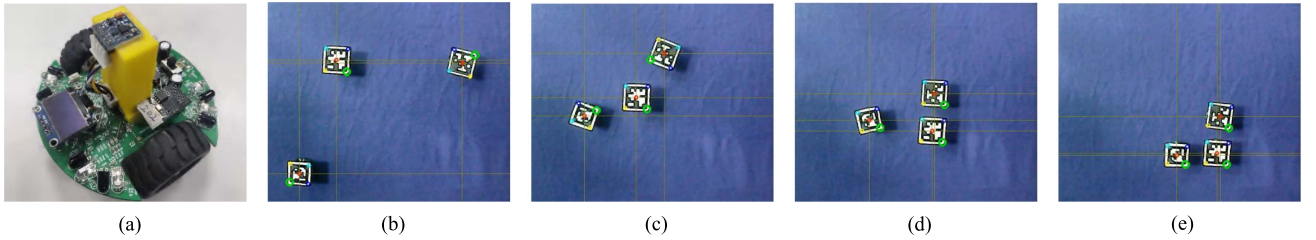


Fig. 5. Robots and their states at different time points. (a) A wheeled robot called MSMR8.0. (b) $t = 0$ s. (c) $t = 7$ s. (d) $t = 13$ s. (e) $t = 17$ s.

named MSMR8.0. The STM32F103VBT6 is used as the main control chip of each robot, the communication module ESP8266 is used to realize WiFi communication between the robots, and the triaxial magnetometer GY-271 is used to measure the orientation angle of each robot. To obtain the position and more accurate orientation angle of each robot in real-time, a distinct tag is affixed directly above each robot, and the AprilTag visual reference system is employed to record and send the orientation angle and position to the corresponding numbered robot by wireless communication. The robot MSMR8.0 is depicted in Fig. 5(a) and these three robots are randomly placed in a square field within a side length of 1.1 m. These robots arranged in a directed network topology according to Fig. 3 are propelled in the field at a speed of 0.05 m/s. The current orientation angle and position of each robot are continually acquired and control protocol (9) is used for the robots to reach consensus at a control frequency of 50 Hz. To assess the efficacy of the protocol, the robots are started with random initial positions and orientations. The purpose of these experiments is to reach consensus of the robots on the orientation angle and position.

The states of these three wheeled robots at various time points such as $t = 0$ s, $t = 7$ s, $t = 13$ s, and $t = 17$ s are shown from Fig. 5(b) to (e). Initially, the robots are dispersed along the edge of the field and orient in different directions. However, the robots under control protocol (9) continually adjust their positions

along the X and Y axes, as well as their orientation angles, such that the relative distances and angles between them gradually approach the desired values. Overall, these experiments demonstrate that consensus of the robots on the orientation angles and positions can be achieved, which exemplifies the correctness of the theory proposed in this article.

VII. CONCLUSION

Based on a novel nonlinear fast convergence function and an NNESO, a consensus control protocol with disturbance compensation is proposed for MASs with disturbances to reach consensus. This protocol enables MASs over an undirected and connected network topology or a directed network topology with a spanning tree to estimate and compensate for unknown total disturbances, which greatly reduces the impact of disturbances on consensus control of MASs with total disturbances. Under this protocol, the state errors between agents can be less than the theoretical maximum state error after MASs are in steady states. By simulations and actual experiments, it is verified that the performance of the NNESO is superior to that of other ESOs, and the NNESO based MASs with disturbances can reach consensus. In future work, the results of this article will be generalized to formation control and optimal control of MASs with disturbances.

APPENDIX

PROOF OF THE FAST CONVERGENCE OF $scf(s, a, b)$

In this appendix, the convergence speed of $scf(s, a, b)$ and that of $fal(s, \mu, \delta)$ will be compared under $b = \mu$. To compare the convergence speed of the two functions, we can consider the following two systems:

$$\dot{s} = -scf(s, a, b) \quad (52)$$

and

$$\dot{s} = -fal(s, \mu, \delta). \quad (53)$$

The Lyapunov candidate function of system (52) is set to $V_{scf} = s^2$, and its derivative is

$$\dot{V}_{scf} = 2s\dot{s} = -2\eta \left(|s|^{a+1} + a^{\frac{a}{|s|+b}} |s|^{b+1} \right) < 0. \quad (54)$$

Similarly, the Lyapunov candidate function of system (53) is set as $V_{fal} = s^2$, and its derivative is

$$\dot{V}_{fal} = 2s\dot{s} = \begin{cases} -|s|^{\mu+1} < 0, & |s| > \delta, \\ -\frac{s^2}{\delta^{1-\mu}} < 0, & |s| \leq \delta. \end{cases} \quad (55)$$

From the stability analysis, it can be concluded that both systems (52) and (53) are able to converge asymptotically to 0 at the speeds of $scf(s, a, b)$ and $fal(s, \mu, \delta)$, respectively. Therefore, their rates of convergence will be distinguished by comparing the sizes of $scf(s, a, b)$ and $fal(s, \mu, \delta)$.

First, it is necessary to judge whether they have the same convergence direction or not. Since

$$\begin{aligned} sgn[scf(s, a, b)] &= sgn \left[\eta \left(|s|^a + a^{\frac{a}{|s|+b}} |s|^b \right) sgn(s) \right] \\ &= sgn(\eta s) = sgn(s) \end{aligned} \quad (56)$$

and

$$sgn[fal(s, \mu, \delta)] = \begin{cases} sgn[|s|^\mu sgn(s)] = sgn(s), & |s| > \delta, \\ sgn\left(\frac{s}{\delta^{1-\mu}}\right) = sgn(s), & |s| \leq \delta \end{cases} \quad (57)$$

then $sgn[scf(s, a, b)] = sgn[fal(s, \mu, \delta)]$, which indicates that $scf(s, a, b)$ and $fal(s, \mu, \delta)$ have the same convergence direction.

On the basis of the above research, the convergence speed of the two functions can be compared, and the comparison function is designed as

$$\begin{aligned} E(s) &= |scf(s, a, b)| - |fal(s, \mu, \delta)| \\ &= |scf(s, a, b)| - |fal(s, b, \delta)| \\ &= \begin{cases} E_1(s), & |s| > \delta, \\ E_2(s), & |s| \leq \delta \end{cases} \end{aligned} \quad (58)$$

where

$$\begin{aligned} E_1(s) &= \left| \eta \left(|s|^a + a^{\frac{a}{|s|+b}} |s|^b \right) sgn(s) \right| - |s|^b sgn(s)| \\ &= \eta \left(|s|^a + a^{\frac{a}{|s|+b}} |s|^b \right) - |s|^b \\ &= \eta |s|^a + \left(\eta a^{\frac{a}{|s|+b}} - 1 \right) |s|^b \end{aligned} \quad (59)$$

and

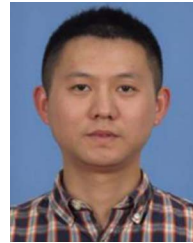
$$\begin{aligned} E_2(s) &= \left| \eta \left(|s|^a + a^{\frac{a}{|s|+b}} |s|^b \right) sgn(s) \right| - \left| \frac{s}{\delta^{1-b}} \right| \\ &= \eta \left(|s|^a + a^{\frac{a}{|s|+b}} |s|^b \right) - \frac{|s|}{\delta^{1-b}} \\ &= \eta |s|^a + \left(\eta a^{\frac{a}{|s|+b}} \delta^{1-b} |s|^{b-1} - 1 \right) \frac{|s|}{\delta^{1-b}}. \end{aligned} \quad (60)$$

Since $b, \delta \in (0, 1)$, $a > 1$, and $\eta \geq 1$, then $E_1(s) > 0$, $E_2(s) > 0$, and $E(s) > 0$, which indicate that the convergence speed of $scf(s, a, b)$ is faster than that of $fal(s, \mu, \delta)$ under the appropriate parameters.

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